

Skill-3

1. To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.

```
ananya@DESKTOP-42I5F93:~$ cd ~
ananya@DESKTOP-42I5F93:~$ mkdir Skill3
mkdir: Skill3: File exists
ananya@DESKTOP-42I5F93:~$ cd Skill3
ananya@DESKTOP-42I5F93:~/Skill3$ nano myshell.c
ananya@DESKTOP-42I5F93:~/Skill3$ gcc -Wall -Wextra -g myshell.c -o myshell
ananya@DESKTOP-42I5F93:~/Skill3$ ./myshell
myshell> hello
Command stored: hello
myshell> world
Command stored: world
myshell> test
Command stored: test
myshell> exit
Shell terminated.
```

```
ananya@DESKTOP-42I5F93:~/Skill3$ valgrind --version
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
ananya@DESKTOP-42I5F93:~/Skill3$ gcc -Wall -Wextra -g myshell.c -o myshell
ananya@DESKTOP-42I5F93:~/Skill3$ valgrind --leak-check=full ./myshell
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
ananya@DESKTOP-42I5F93:~/Skill3$ sudo apt install valgrind
Installing:
  valgrind

Installing dependencies:
  gdb                libdebuginfod-common  libipt2
  libbabeltrace1     libdebuginfod1t64      libsource-highlight-common
  libc6-dbg          libdw1t64              libsource-highlight4t64

Suggested packages:
  gdb-doc  gdbserver  valgrind-mpi  kcache-grind

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 69
  Download size: 27.1 MB
  Space needed: 111 MB / 1024 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod-common all 0.194-4 [12.3 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdw1t64 amd64 0.194-4 [285 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libbabeltrace1 amd64 1.5.11-5build1 [165 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod1t64 amd64 0.194-4 [20.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 libipt2 amd64 2.1.2-3 [49.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight-common all 3.1.9-4.3build2 [64.2 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libsource-highlight4t64 amd64 3.1.9-4.3build2 [273 kB]
```

```
Selecting previously unselected package libdebuginfod1t64:amd64.
Preparing to unpack .../3-libdebuginfod1t64_0.194-4_amd64.deb ...
Unpacking libdebuginfod1t64:amd64 (0.194-4) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../4-libipt2_2.1.2-3_amd64.deb ...
Unpacking libipt2 (2.1.2-3) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../5-libsource-highlight-common_3.1.9-4.3build2_all.deb ...
Unpacking libsource-highlight-common (3.1.9-4.3build2) ...
Selecting previously unselected package libsource-highlight4t64:amd64.
Preparing to unpack .../6-libsource-highlight4t64_3.1.9-4.3build2_amd64.deb ...
Unpacking libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Selecting previously unselected package gdb.
Preparing to unpack .../7-gdb_17.1-2ubuntu1_amd64.deb ...
Unpacking gdb (17.1-2ubuntu1) ...
Selecting previously unselected package libc6-dbg:amd64.
Preparing to unpack .../8-libc6-dbg_2.43-2ubuntu2.3_amd64.deb ...
Unpacking libc6-dbg:amd64 (2.43-2ubuntu2.3) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../9-valgrind_1%3a3.26.0-0ubuntu1_amd64.deb ...
Unpacking valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod-common (0.194-4) ...
Setting up libsource-highlight-common (3.1.9-4.3build2) ...
Setting up libc6-dbg:amd64 (2.43-2ubuntu2.3) ...
Setting up libdw1t64:amd64 (0.194-4) ...
Setting up libipt2 (2.1.2-3) ...
Setting up libbabeltrace1:amd64 (1.5.11-5build1) ...
Setting up valgrind (1:3.26.0-0ubuntu1) ...
Setting up libdebuginfod1t64:amd64 (0.194-4) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build2) ...
Setting up gdb (17.1-2ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
ananya@DESKTOP-42I5F93:~/Skill3$ valgrind --version
valgrind-3.26.0
ananya@DESKTOP-42I5F93:~/Skill3$ valgrind --leak-check=full ./myshell
==1205== Memcheck, a memory error detector
==1205== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==1205== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==1205== Command: ./myshell
==1205==
```

```
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valgrind-3.26.0
ananya@DESKTOP-42I5F93:~/Skill3$ valgrind --leak-check=full ./myshell
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==1205== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==1205== Command: ./myshell
==1205==
myshell> hello
Command stored: hello
myshell> world
Command stored: world
myshell> exit

Shell terminated.
==1205==
==1205== HEAP SUMMARY:
==1205==    in use at exit: 0 bytes in 0 blocks
==1205==    total heap usage: 6 allocs, 6 frees, 1,168 bytes allocated
==1205==
==1205== All heap blocks were freed -- no leaks are possible
==1205==
==1205== For lists of detected and suppressed errors, rerun with: -s
==1205== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
ananya@DESKTOP-42I5F93:~/Skill3$
```